Emma Torres Chickles

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Summary

ML researcher working on generative priors, inverse problems, and representation learning for scientific data at survey scale. My work centers on one question: when a model reconstructs something we cannot directly observe, how much of the answer is measurement and how much is the prior? I build the diagnostics that separate the two, and the visualizations that make the distinction legible.

Experience

PhD Researcher — MIT Department of Physics Aug 2021–Present

Generative priors and inverse problems, representation learning, and large-scale time-series ML

- Built a **score-based diffusion prior** over 2-D physical fields and coupled it to a known linear forward operator for posterior sampling; benchmarked four samplers (**DPS**, **DAPS**, **PnP-DM**, Tikhonov) and showed reconstruction quality is **sampler-agnostic but prior-dominated** (fidelity 0.74–0.78 across all four).
- Quantified **prior-induced error** that posterior uncertainty cannot express: two equally defensible learned priors agree to 21% on the data-constrained subspace while differing 61% on the operator's null space, and single-run error bars understate that disagreement by **1.6–2.4**× over a third to half of the reconstruction.
- Developed self-supervised continuous-time transformer and contrastive models for irregularly sampled time series, with diagnostics separating genuine temporal structure from class-template matching — showing strong downstream performance can be driven by template matching rather than genuine temporal understanding (ICML workshop, 2026).
- Benchmarked pretrained and from-scratch time-series foundation models (**Chronos-T5**, **MOMENT**, **PatchTST**) under matched-scale controls to isolate the effect of cross-domain pretraining on downstream performance.
- Built a streaming pipeline over **11 TB** of HDF5, indexing **1.4B+ sources** for self-supervised training on irregular, heteroscedastic real-world time series.
- Deployed a GPU-accelerated search across **16 NVIDIA A100s** via Slurm job arrays, processing **1.4M+ time series** and cutting end-to-end runtime from months to ∼**6 hours**.
- Led discovery of two compact binary systems (ApJ 2025, 2026) — designed the search strategy, coordinated follow-up observations, and integrated heterogeneous datasets across varying cadences and noise regimes.

Research Assistant — Wellesley College Jun 2018–May 2021

Representation learning and computer vision

Trained convolutional autoencoders on ∼200k light curves with hyperparameter-sweep infrastructure for unsupervised representation learning; applied transfer learning with pretrained CNNs to classify terrain in orbital imagery (**92%** on ∼80k labeled maps).

Selected Projects

- **Starspot mapping as an inverse problem** — extended **InverseBench** (ICLR 2025) with a new scientific inverse problem: reconstructing a 4,608-parameter surface map from a single brightness time series on a real survey cadence. Showed the data constrain only **19 of 4,608** parameters (99.6% null space), so such reconstructions are overwhelmingly prior-driven, and that a mismatched prior yields confident structure *anti-correlated* with truth. **Best Visualization**, IAIFI Summer School hackathon (2026).
- **Interactive embedding explorer** — browser tool for inspecting learned representations, letting users compare how embedding spaces organize real physical structure versus survey systematics.

Selected Publications

- **Chickles, E.**, et al. *ICML AI4Physics Workshop* (2026, accepted) — Template matching, not time learning: a diagnostic for self-supervised light curve encoders
- **Chickles, E.**, et al. *ApJ* (2026) — An eclipsing 8.56-minute orbital period mass-transferring binary
- **Chickles, E.**, et al. *ApJ* (2025) — A gravitational-wave–detectable Type Ia supernova progenitor

Skills

Languages & tools: Python (PyTorch, NumPy, SciPy, Pandas, scikit-learn), Git, Slurm, LaTeX

ML: score-based/diffusion generative priors; Bayesian inverse problems and posterior sampling; uncertainty quantification and calibration; self-supervised & contrastive representation learning; transformers for irregularly sampled sequences; time-series foundation models

Scale: distributed multi-GPU training (PyTorch DDP), multi-node job arrays, TB-scale data pipelines (HDF5)

Awards

Best Visualization, IAIFI Summer School hackathon (2026) | MIT School of Science Community Service Fellowship (2026) | Principal Investigator, Magellan Telescope observing time, multiple competitive allocations (2022–2025) | Sarah Frances Whiting Medal for Achievement in Astronomy (Wellesley, 2020)

Leadership

Mentored 3 researchers (high school to graduate level), 2 projects yielding co-authored manuscripts | President, MIT Physics Graduate Student Council (2023–24) | President, MIT Physics Graduate LGBTQ+ Organization (2024–present)

Education

MIT — Ph.D. Physics, expected May 2027 | **Wellesley College** — B.A. Physics & Astronomy, *magna cum laude*, 2021